

master acts as a certificate authority, and must generate its own certificates which are used to sign agent certificate requests for authentication.

When you run the Puppet agent the first time, it will generate an SSL certificate and send a signing request to the Puppet master. The client can communicate and be controlled by the Puppet master server only after the agent's certificate is signed. All the agent certificates must be signed by the master server.

Before creating a certificate, we have to edit the `/etc/puppet/puppet.conf` file and append the following lines in the `[main]` section:

```
certname = puppet
dns_alt_names = puppet.master.com
```

To generate a certificate on the master, use the command shown below:

```
#puppet master --no-daemonize
```

To list all unsigned certificate requests, use the following command:

```
#puppet cert list
```

Now the master is ready. Let's now install agents on client servers. After you download the Puppet agent, install it as follows:

```
#yum install puppet
```

To install the Puppet agent, use the following command:

```
#vi /etc/default/puppet
modify START=yes
```

To add the master server entry in the client, use the command given below:

```
#vi /etc/puppet/puppet.conf
```

Delete the `[master]` section. Add the following line in the `[agent]` section:

```
Server = puppet.master.com
```

Start the Puppet agent, as follows:

```
#service puppet start
```

Once you start the agent, it will create and send the certificate to the master server. Once the certificate is

authenticated, the client can communicate with the master. So log in to the master server to authenticate the client certification.

To list the certificates received, run the following commands from the master server:

```
#puppet cert list This will list the Agent FQDN name
#puppet cert sign <FQDN>
```

In case you have many certificates from multiple clients, you can use 'all' instead of `<FQDN>` to certify all clients. After signing, the client can be controlled by the master server. In any case, if you want to remove the controls from the master, you can delete the certificate by using the following code:

```
#puppet cert clean <FQDN>
```

Repeat the above steps to set up the agent for all client servers and once done, we are good to go—to test Puppet.

How Puppet works

Once you have installed the agent, Puppet collects information about each client by using *Facter*.

If you run the *facter* command in an agent, you will get an output similar to the one shown below.

```
#facter
<output>
login as: root
root@192.168.0.102's password:
Last login: Wed Sep  7 14:25:50 2016
aio_agent_version => 1.5.2
augeas => {
  version => "1.4.0"
}
disks => {
  sda => {
    model => "VBOX HARDDISK",
    size => "20.00 GiB",
    size_bytes => 21474836480,
    vendor => "ATA"
  }
}
dmi => {
  bios => {
    release_date => "12/01/2006",
    <Truncated>
```

By default, the Puppet agent connects with the master server every 30 minutes and shares the status information with the master. In parallel, it will pull the data about the desired status of defined services from the master and take appropriate action to meet the desired status.